

Content Agent Demo

3-day MVP build · cost-efficient stack · multi-platform output · sales-pitch ready

Timeline 3 days	Infra spend < \$30 / mo	Platforms 4 (LI · YT · IG · X)	Demo target < 3 min cold run
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Executive Summary

This plan delivers a working MVP of an AI content agent for sales-pitch demos in three focused build days. The agent takes a brand persona, runs its own research, develops a content strategy, and produces ready-to-publish multi-platform output — including an AI avatar presenter in the brand voice. The MVP uses a deliberately cost-efficient stack so infrastructure spend during the build and ongoing demos stays under \$30/month. The architecture is designed so cinematic-quality avatar and visuals can be swapped in post-validation without restructuring the agent.

The sales close moment is the **four-platform reveal**: prospects see their brand's content rendered side-by-side in LinkedIn, YouTube Shorts, Instagram Reels, and X formats — each in its native aspect ratio, scripted and visualized by the agent in under three minutes.

Demo Vision

The agent demonstrates four capabilities live, each streamed visibly to the UI so the prospect watches it *think*:

- **Researches** trending topics, competitor angles, and audience pain points in the prospect's niche
- **Strategizes** a content plan with pillars, formats, and a weekly cadence
- **Writes** platform-tailored copy in the brand voice — different hook, length, and CTA discipline per channel
- **Renders** an AI avatar presenter delivering the script, composed into each platform's native format

Pre-loaded brand personas let the demo skip cold-start setup. Pre-rendered avatar segments preserve momentum during live walkthroughs.

Agent Architecture

A coordinator agent orchestrates four specialist sub-agents. Each streams its output via server-sent events to the UI — the visible "agent at work" effect is critical to the sales narrative.

Sub-agent	Role	Model
Researcher	Pulls trending topics, competitor content, audience pain points via Tavily; summarizes findings.	Haiku 4.5
Strategist	Proposes content pillars, weekly plan, and per-platform format mix. Structured JSON output.	Sonnet 4.6
Copywriters	Four parallel prompts — one per platform — each with channel-specific voice, length, and CTA rules.	Sonnet 4.6
Art Director	Picks thumbnails, drives avatar render, oversees Remotion composition per platform.	Haiku 4.5

Cost-Efficient Stack

Each component below is chosen for cheapest credible MVP. Components marked **swap-up** are direct upgrade paths to cinematic-tier post-MVP without touching the agent architecture.

Component	MVP choice	Why	Est. cost
LLM — strategy & copy	Claude Sonnet 4.6	Quality where it matters most — strategy and final-draft copy	pay-per-token
LLM — routing & summary	Claude Haiku 4.5	5–10× cheaper for research summarization and simple classification	pay-per-token
Research	Tavily	\$1 per 1k searches; fastest API to set up	< \$5 / mo
Avatar (swap-up)	HeyGen basic API or D-ID	Step down from cinematic; free tier covers full demo cycle	\$0 (free tier)
Voice	ElevenLabs free / starter	10k chars/mo free is enough for demo volume; voice cloning deferred	\$0 – \$5 / mo
Images (swap-up)	Flux Schnell on Replicate	~\$0.003 per image; quality fine for thumbnails and b-roll stills	< \$5 / mo
Video composition	Remotion	Open source; programmatic per-platform reformat (9:16, 16:9, 1:1)	\$0
Frontend	Next.js 14 + shadcn + Framer Motion	Polished defaults; SSE-friendly; one stack for FE + BE	\$0
Backend	Next.js API routes	No separate Python service to maintain; Claude Agent SDK in TS	\$0
DB / Auth / Storage	Supabase free tier	Postgres + auth + storage + realtime in one box	\$0
Hosting	Vercel hobby	Fits MVP load; instant deploy from Git	\$0

Deliberately deferred to post-MVP:

- Argil or HeyGen Studio for cinematic-quality avatar (architecture unchanged when swapped in)
- Voice cloning per prospect (one-time onboarding step)
- Exa for semantic research; Apify for competitor social scraping
- Real social posting integrations (until a prospect commits)

Day 1 — Foundation + Research & Strategy

Goal: end the day with a working app shell where the Researcher and Strategist sub-agents stream output to the UI for a pre-loaded brand persona.

Morning (≈ 4 hrs)

- Next.js 14 project init — shadcn, Tailwind, Framer Motion, Claude Agent SDK
- Supabase project; schema for *brand_profiles*, *content_runs*, *generated_assets*
- Auth bypass for demo (single shared session — skip the login friction)
- Vercel deploy pipeline; .env wiring for Anthropic, Tavily, Supabase
- Seed two demo personas: "B2B SaaS founder" and "Fitness coach"

Afternoon (≈ 4 hrs)

- **Researcher** sub-agent: Tavily integration → Haiku 4.5 summarization → structured findings
- **Strategist** sub-agent: Sonnet 4.6 with structured JSON output (pillars + weekly plan)
- SSE streaming wrapper to push agent state to the UI as it works
- Three-column UI shell: brand picker (left), agent activity log (center), output preview (right)

END-OF-DAY MILESTONE

Pick a persona → click Run → watch research scroll in real time → see the strategy materialize as structured output.

Day 2 — Copy, Voice, Avatar

Goal: end the day with platform-specific copy generated in parallel, plus voiceover and avatar video render for one full content piece.

Morning (≈ 4 hrs)

- Four **Copywriter** sub-agents — LinkedIn long-form, YouTube Shorts script, Instagram Reels caption + script, X thread — each with channel-specific system prompt
- Run all four in parallel; persist outputs with platform metadata
- Tabbed preview pane in the UI — one tab per platform
- Brand-voice consistency check (single Sonnet pass after parallel writes)

Afternoon (≈ 4 hrs)

- **Voice generation:** ElevenLabs API from the final script (default voice — cloning deferred)
- **Avatar render:** HeyGen or D-ID API — send script + voiceover, store returned video URL
- Store all rendered video in Supabase storage; *cache by script hash* so repeat runs are instant
- **Thumbnail generation:** Flux Schnell via Replicate

END-OF-DAY MILESTONE

One full content piece for one persona now has: script (all 4 platforms), voiceover, avatar video, and thumbnails — all accessible from the UI.

Day 3 — Composition, Polish, Demo Prep

Goal: end the day demo-ready — all four platforms reformatted to native aspect ratios, the reveal moment animated, the full demo flow rehearsed and pre-rendered.

Morning (≈ 4 hrs)

- Remotion templates: 9:16 (Shorts / Reels), 16:9 (LinkedIn / YT long), 1:1 (IG feed)
- Composition pipeline: avatar video + burned captions + branded intro/outro per platform
- Render each platform variant; store in Supabase keyed to the content run

Afternoon (≈ 3 hrs)

- **Four-platform reveal UI:** native aspect ratios, side-by-side, animated entry — this is the close
- Framer Motion polish on agent log streaming (subtle scroll, fade-in for messages)
- Loading-state choreography that masks the few unavoidable real waits

Final hour

- Dry-run the demo end to end — target sub-3-minute cold run
- Pre-render every asset for both demo personas so live calls don't depend on slow APIs
- Backup screen recordings of each step in case of network failure on a call

END-OF-DAY MILESTONE

*Run the full demo cold in under 3 minutes. End on the four-platform reveal as the close.
Backup recordings ready in case of network issues.*

Risks & Mitigations

Risk	Mitigation
Avatar render latency unpredictable (30s – 5min)	Pre-render all demo persona assets; cache by script hash; keep backup screen recording for live demos.
SSE streaming UI complexity (buffering, reconnects)	Keep payloads small; emit checkpoint events; reserve a half-day buffer late on Day 1 if needed.
Remotion composition edge cases (caption overflow, aspect bleed)	Build and validate one template fully before duplicating to other aspect ratios.
Free-tier rate limits (Tavily, Replicate, HeyGen)	Cache aggressively; mock expensive calls during dev; warm caches before demo calls.
Brand-voice drift across four parallel copy agents	Single Sonnet voice-consistency pass after parallel writes; share style guide in every prompt.
Scope creep — "can we add platform X / feature Y"	Lock the 4 platforms + 2 personas as the MVP fence; track everything else as post-MVP.

Path from MVP to Cinematic

The MVP architecture is designed so each visual-quality lever can be raised independently once we have prospect validation:

- **Avatar**: swap HeyGen basic → Argil or HeyGen Studio for cinematic-grade rendering (same API surface, no agent changes)
- **Voice**: add ElevenLabs voice cloning per prospect — one-time 60-second sample at onboarding
- **Research depth**: add Exa for semantic search, Apify for competitor social scraping
- **Distribution**: integrate posting APIs (LinkedIn, Buffer/Ayrshare for the rest) once a prospect commits
- **Brand intake**: replace seed personas with live brand-profile ingestion from a URL or uploaded docs

The whole point of the MVP fence is to validate the agent narrative cheaply. Every deferred item above is a Day-4-and-beyond upgrade, not a rewrite.